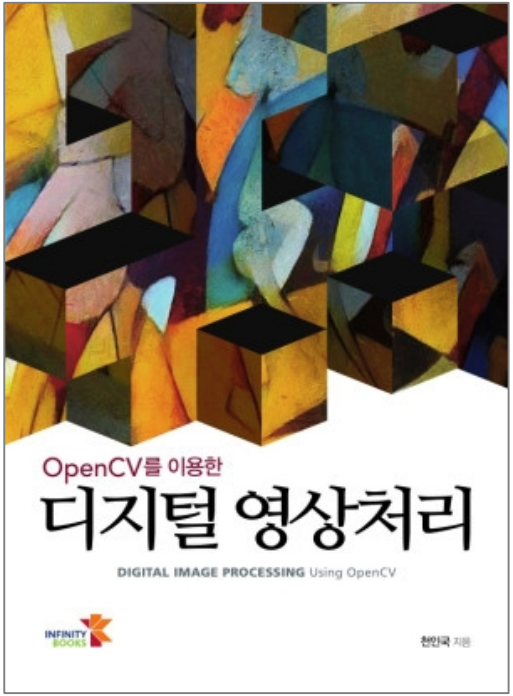
다수이종센서융합SW설계

Sensor Fusion

intRo-Lab.

청주대학교 무인항공기학과
김상현 교수 (k3special@cju.ac.kr)

강의교재 1



INFINITY BOOKS

오픈채팅방

- 링크: https://www.intro-lab.com/lecture/

OpenCV 예시

◆ 다운로드 링크

https://www.intro-lab.com/lecture/g01/

Lecture Note: 다수이종센서융합SW설계(g01)

다수이종센서융합SW설계_01주차.pdf

VSCode_with_OpenCV_intRoLab.zip

opencv_test_2026_03_03.zip

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Visual Studio Code with OpenCV

- 주의: 본 프로그램은 intRo-Lab. 연구실 저작권 자산이므로 본 강의 외 사용 금지
- 다운로드 링크

https://www.intro-lab.com/lecture/g01/

Lecture Note: 다수이종센서융합SW설계(g01)

다수이종센서융합SW설계_01주차.pdf

VSCode_with_OpenCV_intRoLab.zip

opencv_test_2026_03_03.zip

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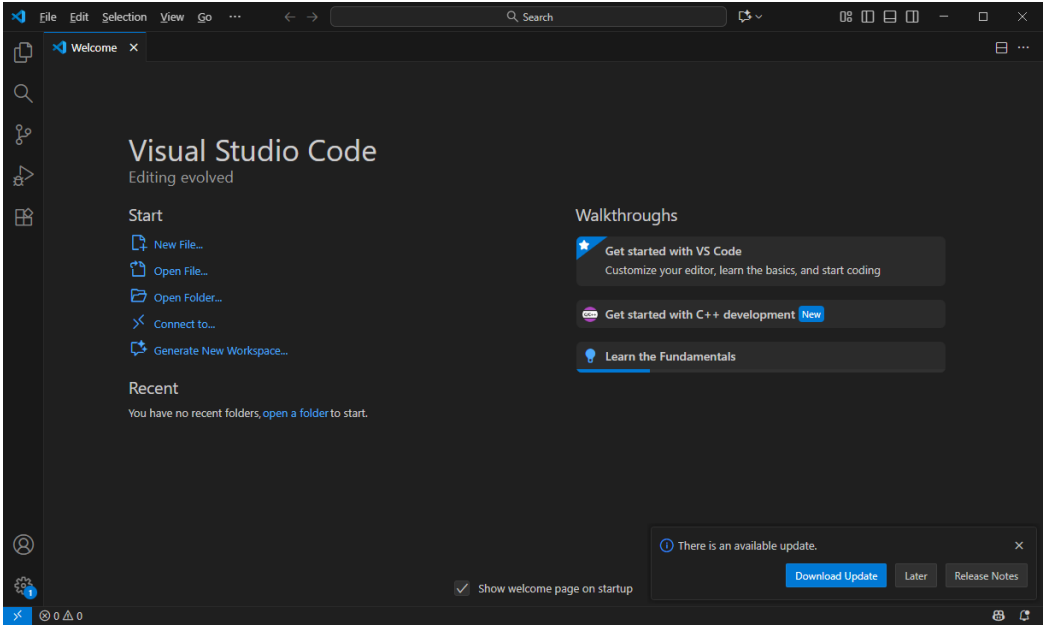
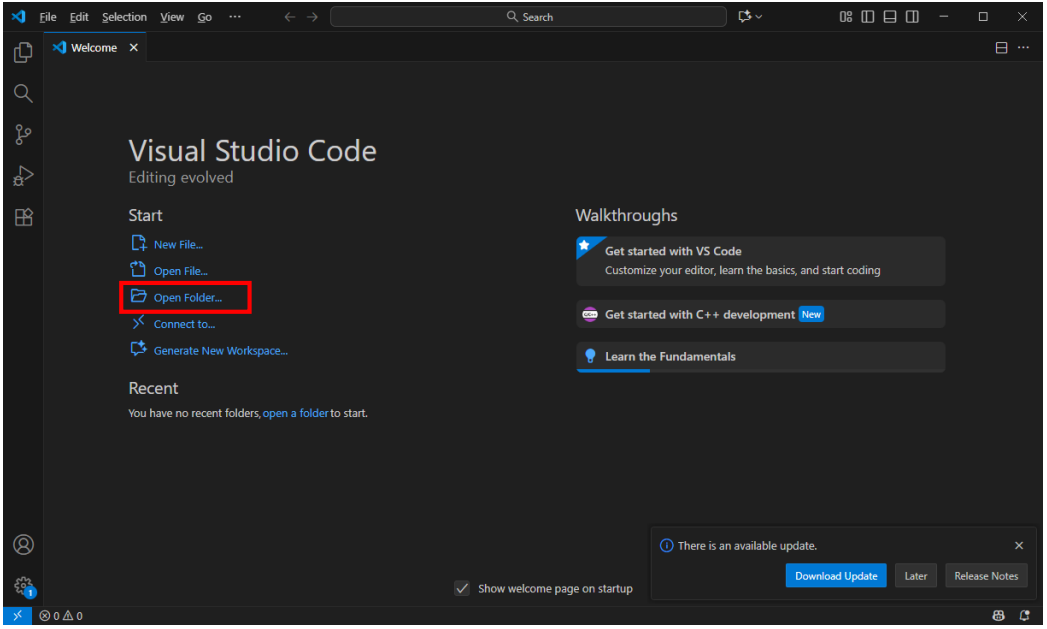
◆ 첫 실행 화면


File
Edit
Selection
View
Go
Search
Welcome
Visual Studio Code
Editing evolved
Start
New File...
Open File...
Open Folder...
Connect to...
Generate New Workspace...
Recent
You have no recent folders, open a folder to start.
Walkthroughs
Get started with VS Code
Customize your editor, learn the basics, and start coding
Get started with C++ development
New
Learn the Fundamentals
There is an available update.
Download Update
Later
Release Notes
Show welcome page on startup

◆ 예시 프로젝트 폴더 열기


File
Edit
Selection
View
Go
Search
Welcome
Visual Studio Code
Editing evolved
Start
New File...
Open File...
Open Folder...
Connect to...
Generate New Workspace...
Recent
You have no recent folders, open a folder to start.
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Get started with VS Code
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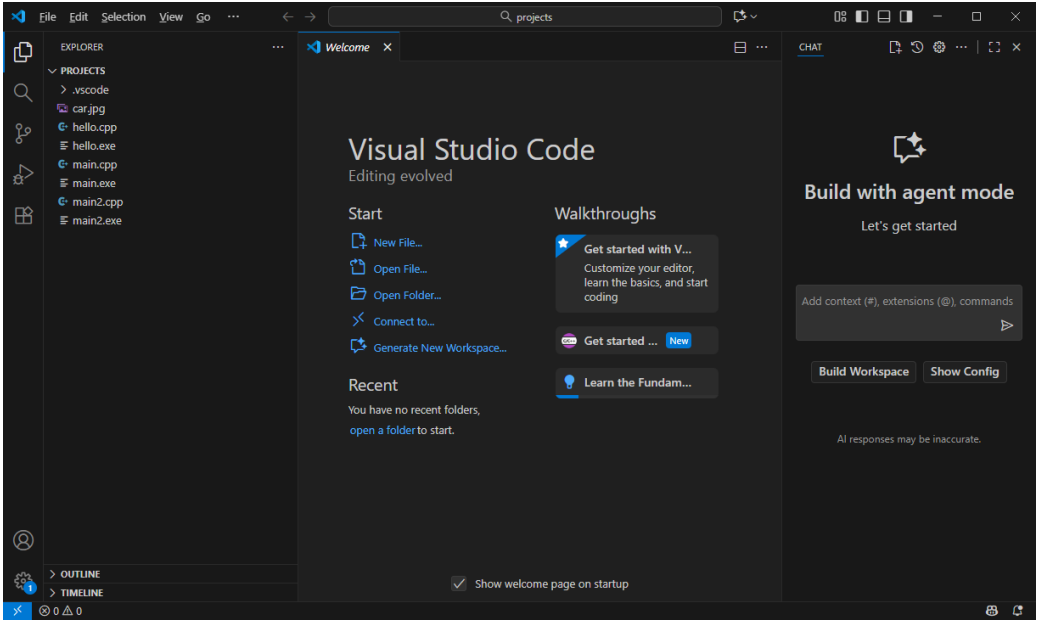
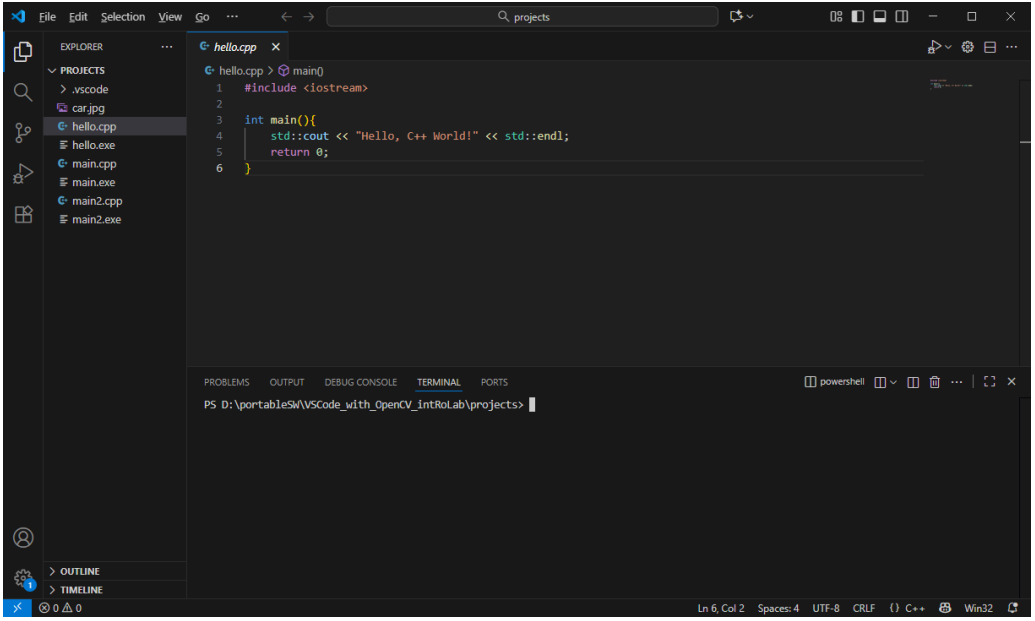
Learn the Fundamentals
There is an available update.
Download Update
Later
Release Notes
Show welcome page on startup

◆ 예시 프로젝트 폴더 열기


File
Edit
Selection
View
Go
projects
EXPLORER
PROJECTS
.vscode
car.jpg
hello.cpp
hello.exe
main.cpp
main.exe
main2.cpp
main2.exe
OUTLINE
TIMELINE
Welcome
Visual Studio Code
Editing evolved
Start
New File...
Open File...
Open Folder...
Connect to...
Generate New Workspace...
Recent
You have no recent folders,
open a folder to start.
Walkthroughs
Get started with V...
Customize your editor, learn the basics, and start coding
Get started ...
New
Learn the Fundam...
Show welcome page on startup
CHAT
Build with agent mode
Let's get started
Add context (#), extensions (@), commands
Build Workspace
Show Config
AI responses may be inaccurate.

◆ 터미널 창 활성


File Edit Selection View Go
projects
EXPLORER
PROJECTS
.vscode
car.jpg
hello.cpp
hello.exe
main.cpp
main.exe
main2.cpp
main2.exe
hello.cpp
hello.cpp > main()
#include <iostream>
int main(){
std::cout << "Hello, C++ World!" << std::endl;
return 0;
}
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
powershell
PS D:\portableSW\VSCode_with_OpenCV_intRoLab\projects>
OUTLINE
TIMELINE
Ln 6, Col 2 Spaces: 4 UTF-8 CRLF {} C++ Win32

영상처리 개요

영상 처리란?

- 디지털 카메라를 통하여 영상을 전자적으로 얻은 후, 컴퓨터로 영상을 처리하여 원하는 출력 영상을 얻는 기술 분야
- 예) 포토샵

영상 출처: 어도비사 홈페이지

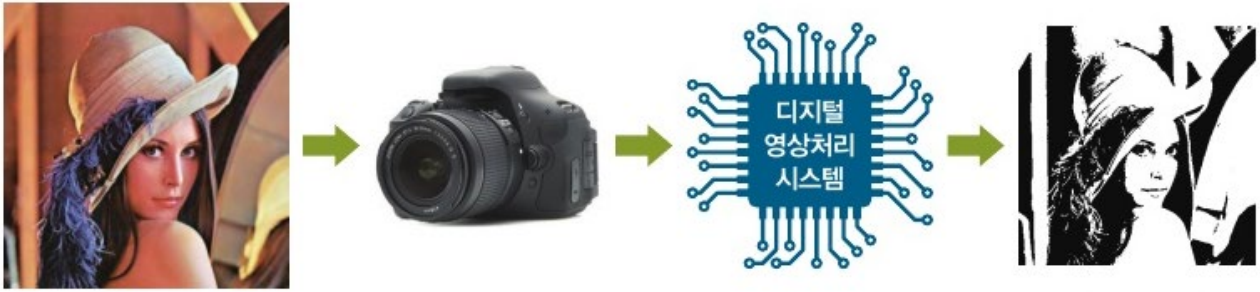
디지털 영상

◆ 영상은 수학적으로는 2차원 함수 $f(x, y)$로 정의될 수 있으며, f의 값이 그 점에서의 영상의 밝기

20	99	50	12	17	60	90	65	76	88
16	45	78	89	34	12	34	55	66	88
...									

$f(x, y)$

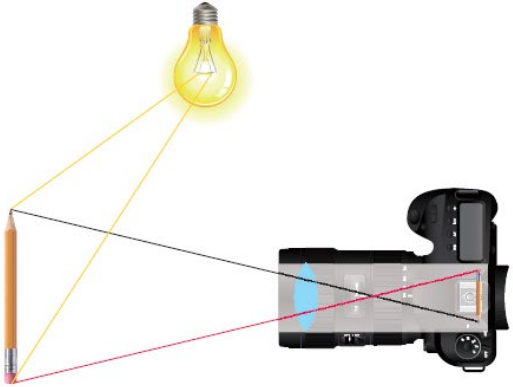
영상처리시스템

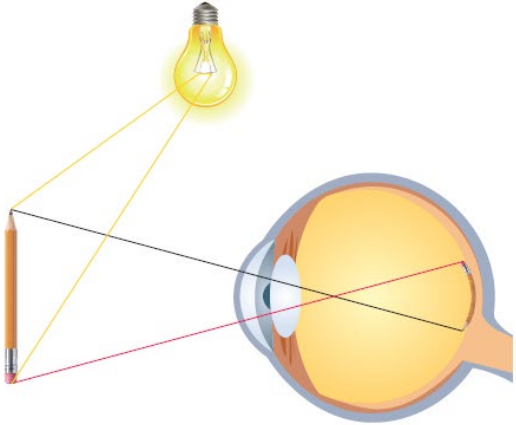


실제 세계 카메라 처리된 영상

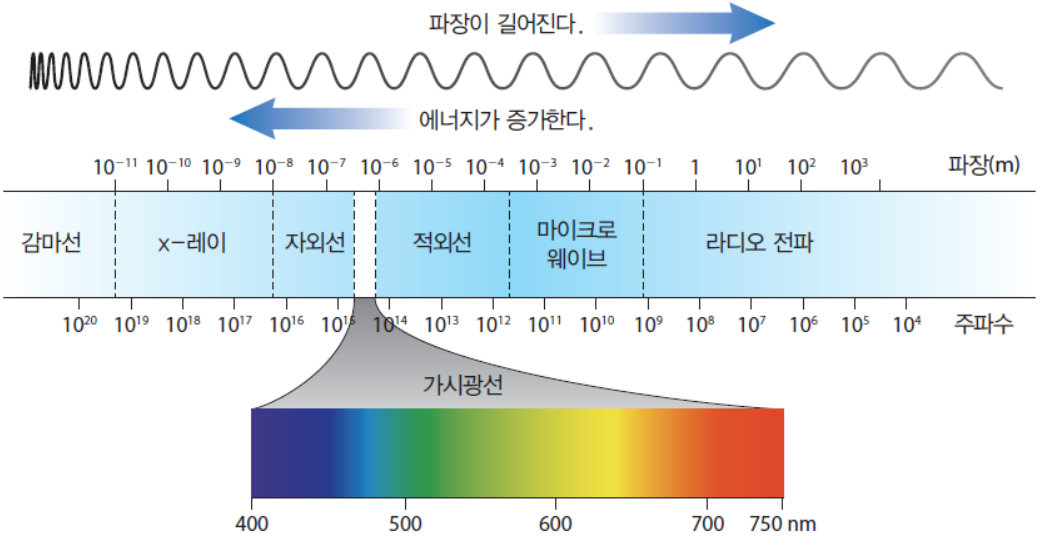
영상의 획득

카메라



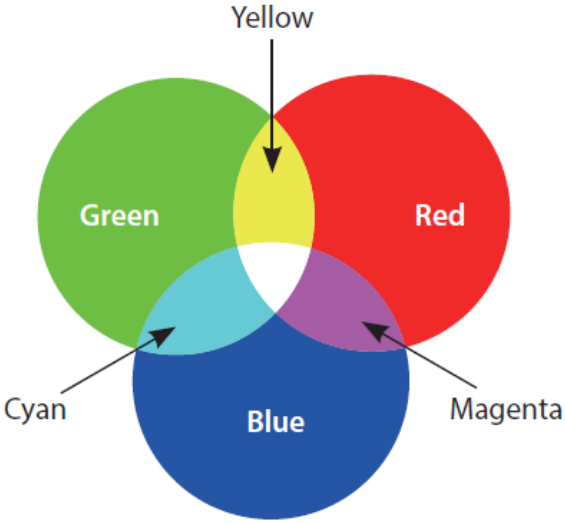


컬러

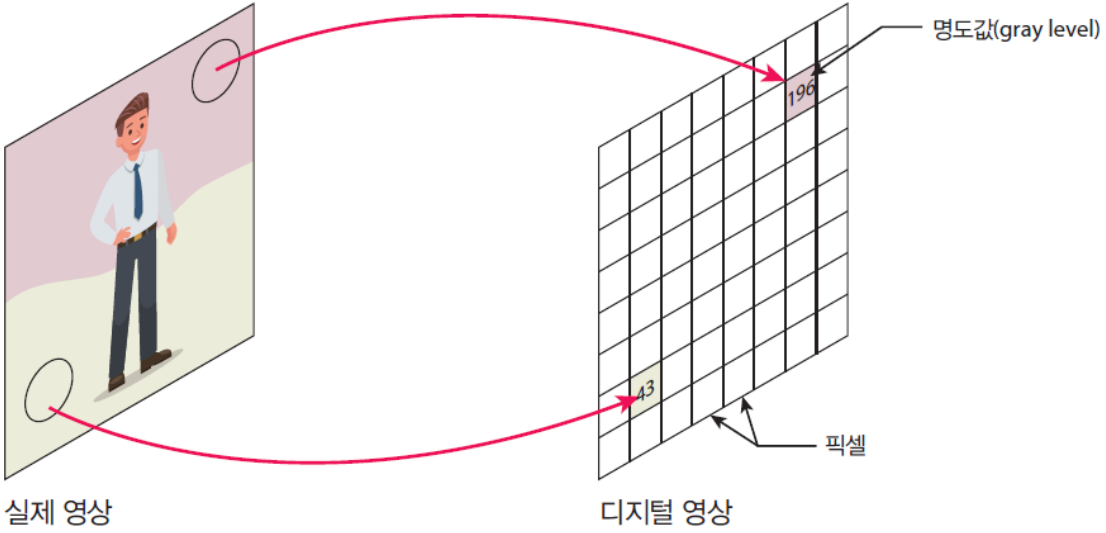

파장이 길어진다.
에너지가 증가한다.
10^{-11} 10^{-10} 10^{-9} 10^{-8} 10^{-7} 10^{-6} 10^{-5} 10^{-4} 10^{-3} 10^{-2} 10^{-1} 1 10^{1} 10^{2} 10^{3}
파장(m)
감마선
x-레이
자외선
적외선
마이크로 웨이브
라디오 전파
10^{20} 10^{19} 10^{18} 10^{17} 10^{16} 10^{15} 10^{14} 10^{13} 10^{12} 10^{11} 10^{10} 10^{9} 10^{8} 10^{7} 10^{6} 10^{5} 10^{4}
주파수
가시광선
400
500
600
700
750 nm

기본 색상

◆ 19세기 후반에 물리학자 맥스웰은 컬러 영상은 3개의 기본 컬러를 이용하여 생성될 수 있음을 보였다.



디지털 영상의 생성

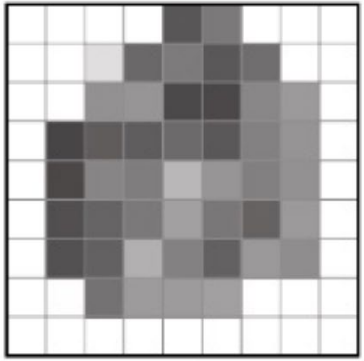


표본화와 양자화

아날로그 영상

(a)

표본화



(b)

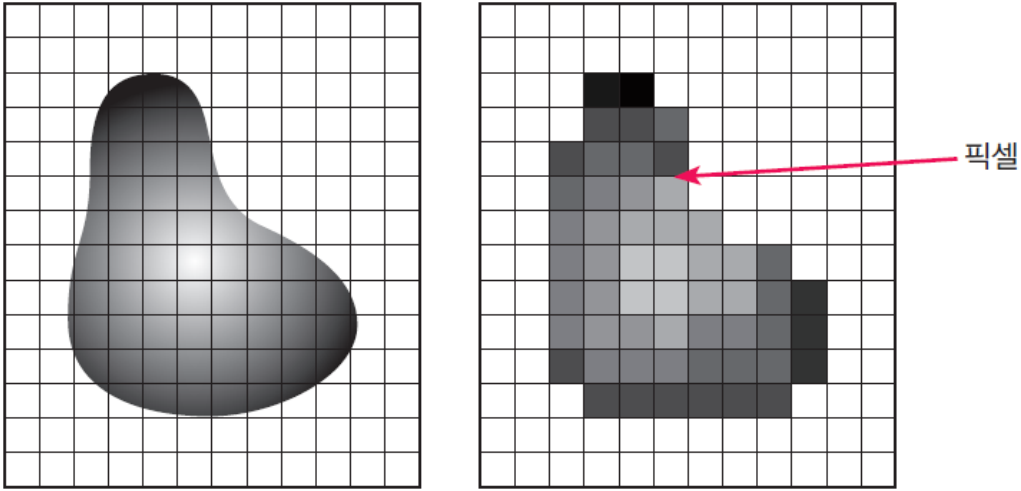
양자화

255	255	255	255	98	139	255	255	255
255	255	230	119	140	101	126	255	255
255	255	161	166	81	80	151	171	255
255	75	105	107	120	100	147	165	255
255	76	150	141	197	165	145	162	255
255	89	113	140	175	136	111	171	255
255	90	111	191	174	111	169	158	255
255	255	131	172	172	176	255	255	255
255	255	255	255	255	255	255	255	255

(c)

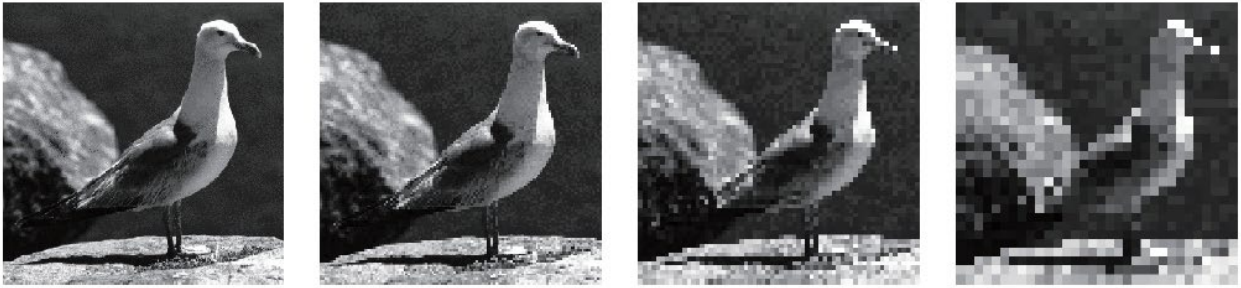
표본화

◆ 공간 영역에서 픽셀의 개수를 제한하는 것이 영상에서의 표본화



표본화

◆ 표본화 == 해상도

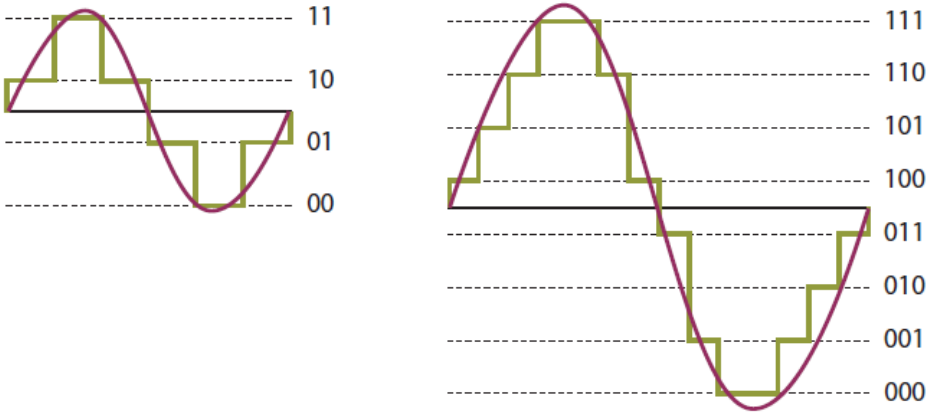


(a) 256×256 영상 (b) 128×128 영상 (c) 64×64 영상 (d) 32×32 영상

[그림 1.9] 해상도에 따른 영상의 변화

양자화

◆ 화소의 값을 정해진 몇 단계의 밝기로 제한하는 과정



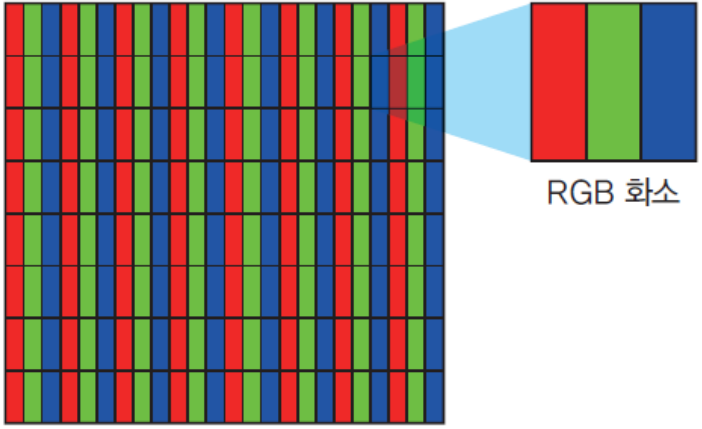
[그림 1.10] 1차원 신호의 양자화

양자화

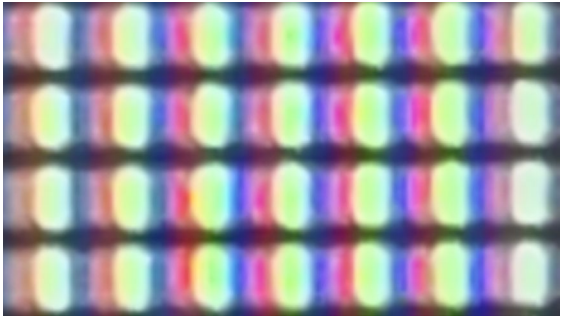
[그림 1.11] 다양한 양자화에 따른 영상의 변화(256, 129, 64, 32, 16, 8, 4, 2)

영상의 종류

- 그레이스케일 영상
- 컬러 영상

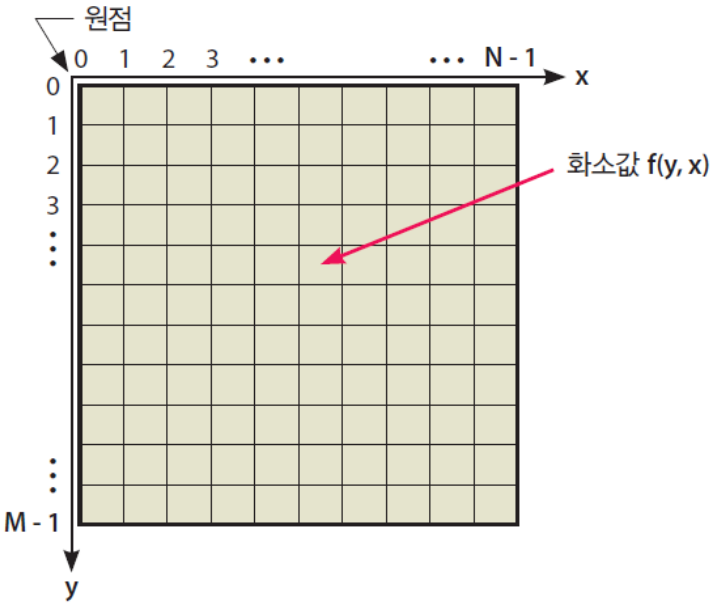


컬러영상

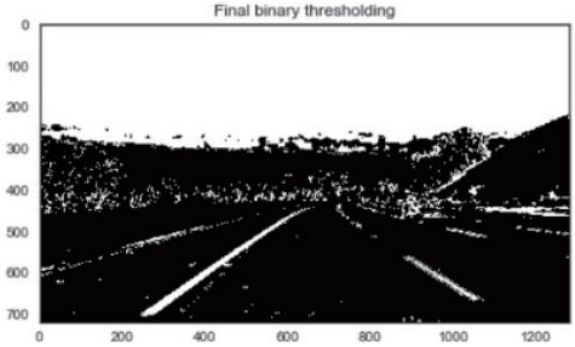


실제 모니터 초근접 촬영

영상의 좌표계



◆ 자율 주행 자동차



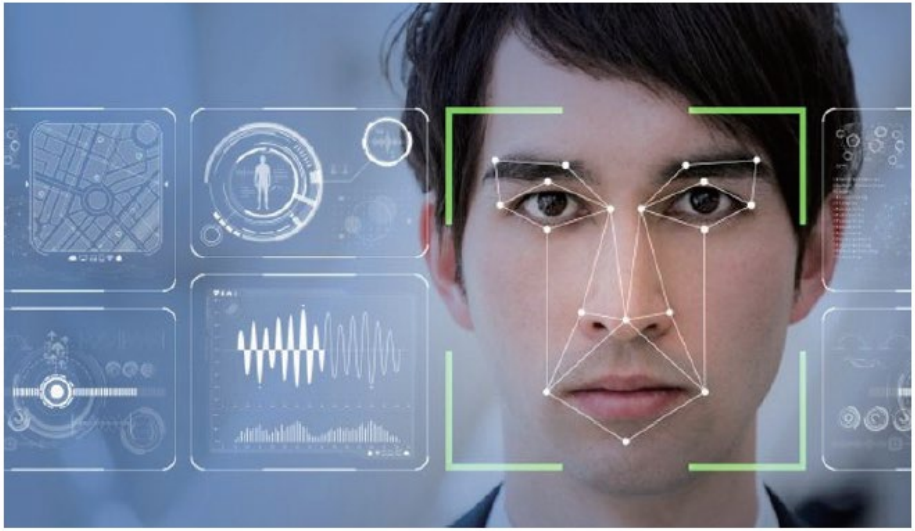
영상출처: https://towardsdatascience.com/

영상처리의 응용

◆ 공장자동화

출처: https://www.automation.com/

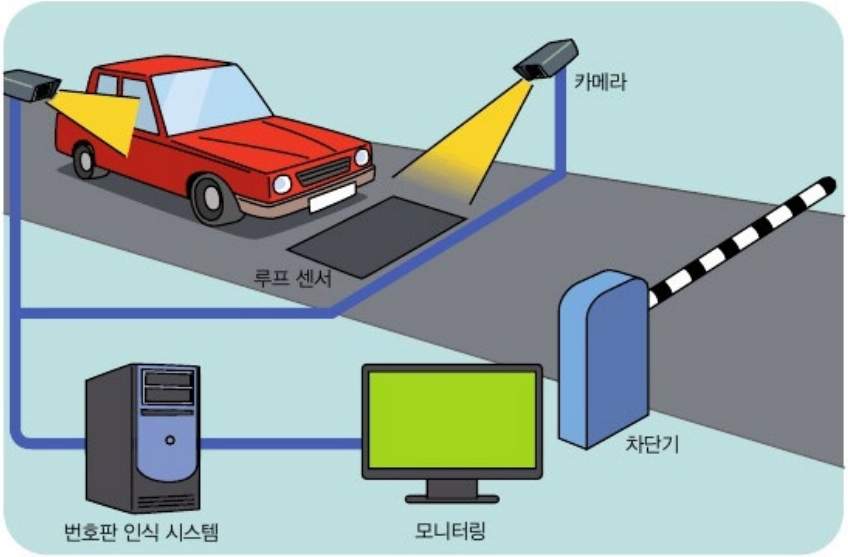
◆ 생체인식 시스템



출처: https://www.techfunnel.com/

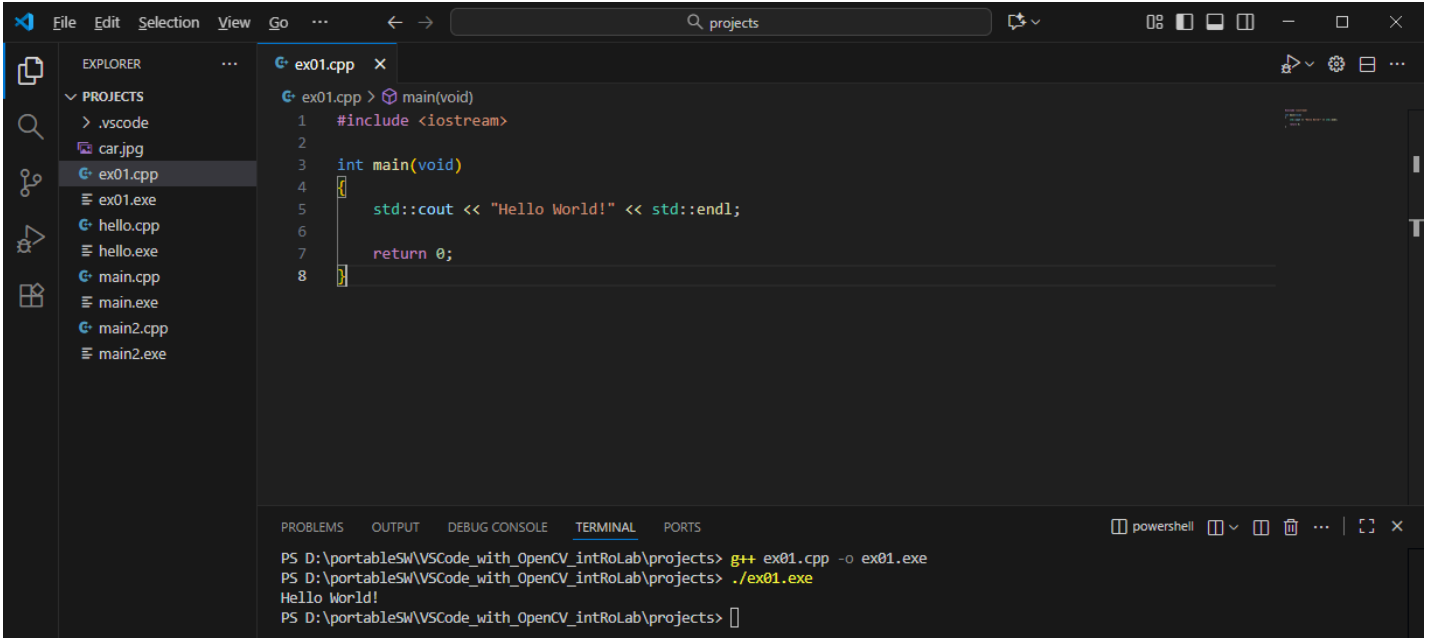
영상처리의 응용

◆ 문서 자동 인식



◆ C++ 언어 기초

https://wikidocs.net/14010



◆ C++ 언어 기초

```
#include <iostream>

int main(void)
{
    std::cout << "Hello World!" << std::endl;

    return 0;
}
```

◆ C++ 언어 기초

File Edit Selection View Go ··· projects

EXPLORER

PROJECTS
- .vscode
- car.jpg
- ex01.cpp
- ex01.exe
- hello.cpp
- hello.exe
- main.cpp
- main.exe
- main2.cpp
- main2.exe

ex01.cpp > main(void)

```cpp
int main(void)
{
    int val1;
    std::cout << "input1: ";
    std::cin >> val1;

    int val2;
    std::cout << "input2: ";
    std::cin >> val2;

    int result = val1 + val2;

    std::cout << "output: " << result << std::endl;

    return 0;
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS D:\portableSW\VSCode_with_OpenCV_intRoLab\projects> g++ ex01.cpp -o ex01.exe
PS D:\portableSW\VSCode_with_OpenCV_intRoLab\projects> ./ex01.exe
input1: 2
input2: 3
output: 5
PS D:\portableSW\VSCode_with_OpenCV_intRoLab\projects>
```

◆ C++ 언어 기초

```cpp
#include <iostream>

int main(void)
{
    int val1;
    std::cout << "input1: ";
    std::cin >> val1;

    int val2;
    std::cout << "input2: ";
    std::cin >> val2;

    int result = val1 + val2;

    std::cout << "output: " << result << std::endl;

    return 0;
}
```

공지 2026.03.10.

◆ 진행 중인 연구 혹은 관심 연구 주제 발표

- ◆ **발표일:** 2026.03.17.
- ◆ **발표시간:** 5분 (간단하게 발표 진행)

- ◆ **제출:** 발표 자료(PPT파일과 PDF 모두 제출)
- ◆ **제출 방법**: 이메일 제출 introlab.office@gmail.com
- ◆ **제출 마감:** 2026.03.16. 23:00 (시간 엄수!)